

BUSINESS CASE

1. TARGET CUSTOMER SEGMENT

- Hospitals and ICUs
- Home care patients (elderly, paralyzed)
- Nursing homes and rehabilitation centers
- Caregivers and family members

2. COMPETITIVE ALTERNATIVES

- Manual nurse call buttons (limited functionality)
- Voice assistants (require internet, not healthcare-specific)
- Patient monitoring systems (focus on vitals, not needs)

3. DIFFERENTIATION

- Combines **button-based and voice-based requests**
- Dedicated buttons for **food, water, restroom**
- **Audio feedback** through dual speakers
- **Low-cost, simple, and easy to use** for bedridden patients

4. ROUGH BOM ESTIMATE

Component	Cost (INR)
Push Buttons	₹200
Speakers(2)	₹400
Power Supply	₹500
PCB BOARD	₹400
Wires & Components	₹200
3D Printed Enclosure	₹1800
Mounting + Misc	₹1500
Total	₹5000

5. DEPLOYMENT PATHWAY /COMMERCIALIZATION IDEA

Deployment Pathway

- **Prototype Development:**
Design and build the initial hardware prototype using SiWG917 module, push buttons, and speaker system. Ensure basic functionality such as input detection and audio output works reliably.
- **Testing & Validation:**
Conduct real-time testing with patients, elderly users, and caregivers to evaluate usability, responsiveness, and accuracy. Collect feedback to identify practical issues.
- **System Enhancement:**
Improve system features by:
 - Integrating voice input recognition
 - Enhancing audio clarity and response speed
 - Reducing power consumption for longer usage
- **Deployment & Implementation:**
Deploy the system in:
 - Hospitals and clinics for patient assistance
 - Home care environments for elderly and disabled individuals.
Ensure easy installation and user-friendly operation.

Commercialization Idea

- Sell as a **low-cost assistive device**
- Target hospitals and home healthcare market
- Provide **after-sales support and maintenance services**
- Collaborate with government and healthcare programs